

Enigma code ultimately had a debilitating impact on German war strategies and helped to end the war.

Cryptography has become a vital part of our lives. Every time we type in a password, pay with a credit card, or use WhatsApp, we are enjoying the benefits of cryptography. Without cryptography, it would be easy for bad actors to steal sensitive information and use it against us. Cryptography makes sure the information can only be used by those for whom it is intended.

Cryptographic

Every transaction recorded in Bitcoin's blockchain must be cryptographically verified to ensure that people trying to send bitcoin actually own the bitcoin they're trying to send.

Cryptography also applies to how groups of transactions are added to Bitcoin's blockchain. Transactions are not added one at a time, but instead in "blocks" that are "chained" together, hence the term blockchain. We will go deeper into the specifics of the process in the proof-of-work section that follows, but for now here's the takeaway: cryptography allows the computers building Bitcoin's blockchain to collaborate in an automated system of mathematical trust. There is no subjectivity as to whether a transaction is confirmed in Bitcoin's blockchain: it's just math. For a deep dive on cryptography, we highly recommend *The Code Book: The Science of Secrecy from Ancient Egypt to Quantum Cryptography* by Simon Singh.